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**NITTE (Deemed to be University)**

(Established under Sec.(3) of UGC Act, 1956)

**First Semester B.Tech (CBCS) Degree Examinations, NOV. - DEC. 2025****25MEC112 – ELEMENTS OF MECHANICAL ENGINEERING**

(For ACT, AER, BTY, CIV, ECE, EEE, MEC, VDT, AID, AIM, CYB, CCE, CSE, CBS, ISE, RAI)

**Duration: 3 Hours]****[Max. Marks: 100****Note:**

**Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five full** questions choosing **One full** question from each Unit.

**PART - A: MULTIPLE CHOICE QUESTIONS****20 Marks**

1. The heating of dry and saturated steam above saturation temperature is known as,  
A) Enthalpy B) Superheating  
C) Super saturation D) Latent heat
2. Which mode of heat transfer occurs through the movement of fluid (liquid or gas) due to differences in temperature and density?  
A) Conduction B) Convection  
C) Radiation D) Insulation
3. What application is associated with steam formed in power plants?  
A) Cooking food B) Generating electricity  
C) Fueling cars D) Air conditioning
4. Surge tanks are found in  
A) Hydroelectric power plant B) Nuclear power plant  
C) Thermal power plant D) Solar power plants
5. Centrifugal pumps are used to transport,  
A) Pressure B) Speed  
C) Power D) Fluid
6. Kaplan turbine is  
A) Axial flow turbine B) Inward flow turbine  
C) Tangential flow turbine D) Mixed flow turbine
7. The following is an example of impulse type of turbine,  
A) Pelton wheel turbine B) Kaplan turbine  
C) Francis turbine D) Parsons turbine
8. In a gas turbine, which type of cycle is used where air is compressed and mixed with fuel before ignition?  
A) Open cycle B) Closed cycle  
C) Rankine cycle D) Otto cycle
9. The ratio of brake power to indicated power is called,  
A) Indicated thermal efficiency B) Relative efficiency  
C) Mechanical efficiency D) Volumetric efficiency
10. Which of the following medium is compressed in a diesel engine cylinder?  
A) Air alone B) Air and fuel  
C) Air and lubrication oil D) Fuel alone
11. What does a Vapor Compression refrigerator use in the form of a refrigerant?  
A) Aqua ammonia B) Freon  
C) Ammonia D) Water
12. Electric vehicles are attracting the world of mobility due to the reason,  
A) Increased pollution by conventional vehicles B) Increasing cost of fossil fuel  
C) Easy to drive D) Light weight

13. To obtain a higher torque, driven gear size should be,  
 A) Smaller than driver B) Larger than driver  
 C) Same as driver D) Do not depend on gear size
14. The teeth on the spur gears with respect to axes of gears are,  
 A) Parallel B) Perpendicular  
 C) Inclined D) Intersecting
15. Solder is an alloy of,  
 A) Lead and Copper B) Zinc and Tin  
 C) Tin and Lead D) Zinc and Copper
16. In cross belt drive the driver and driven pulley always rotates in  
 A) Same direction B) Opposite direction  
 C) Any direction D) None of above
17. Which of the following operation does NOT use a rotating tool?  
 A) Reaming B) Counter-Boring  
 C) Facing D) End milling
18. Which one of the following engineering deals with controlling and intelligence of robots?  
 A) Electrical B) Mechanical  
 C) Computer D) Architecture
19. Producing conical surface on cylindrical work piece is called,  
 A) Taper turning B) Tapping  
 C) Thread cutting D) Reaming
20. Which part of a robot is typically responsible for end-effector tasks and tool manipulation?  
 A) Controller B) Manipulator  
 C) Sensor D) Actuator

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

|                   |                                                                                                                                                                                                                                                                          | 80 Marks |     |     |     |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----|-----|-----|
|                   |                                                                                                                                                                                                                                                                          | Marks    | BT* | CO* | PO* |
| <b>Unit – I</b>   |                                                                                                                                                                                                                                                                          |          |     |     |     |
| 1.                | a) Explain different modes of heat transfer with examples deemed fit.                                                                                                                                                                                                    | 8        | 2   | 1   | 1   |
|                   | b) Explain the working principle of Tidal power plant with a neat sketch.                                                                                                                                                                                                | 8        | 2   | 1   | 1   |
| 2.                | a) State different laws of thermodynamics with suitable examples.                                                                                                                                                                                                        | 8        | 2   | 1   | 1   |
|                   | b) Explain the working principle of Hydel power plant with a neat sketch.                                                                                                                                                                                                | 8        | 2   | 1   | 1   |
| <b>Unit – II</b>  |                                                                                                                                                                                                                                                                          |          |     |     |     |
| 3.                | a) Explain the working Principle of Impulse Steam Turbine (De Laval Turbine) with a sketch.                                                                                                                                                                              | 8        | 2   | 2   | 1   |
|                   | b) Mention the differences between impulse and reaction steam turbines.                                                                                                                                                                                                  | 8        | 2   | 2   | 1   |
| 4.                | a) Explain the working Principle of Closed cycle Gas turbine with a sketch.                                                                                                                                                                                              | 8        | 2   | 2   | 1   |
|                   | b) Explain the working of Impulse water turbine (Pelton wheel) with a sketch.                                                                                                                                                                                            | 8        | 2   | 2   | 1   |
| <b>Unit – III</b> |                                                                                                                                                                                                                                                                          |          |     |     |     |
| 5.                | a) Explain with figure the working principle of a 4-Stroke Petrol engine.                                                                                                                                                                                                | 8        | 2   | 3   | 1   |
|                   | b) Vapor Compression Refrigeration is a four-step cycle system, which provides effective refrigeration. Enumerate the working principle of Vapor Compression Refrigeration system with a neat sketch.                                                                    | 8        | 3   | 3   | 1   |
| 6.                | a) List the Advantages and disadvantages of Electric Vehicles (EVs) and Hybrid vehicles.                                                                                                                                                                                 | 8        | 3   | 3   | 1   |
|                   | b) A four cylinder four-stroke engine running at 1000 rpm develops an indicated power of 15 kW. The mean effective pressure is $5 \times 10 \text{ N/m}^2$ . Find the diameter of the cylinder and the stroke of the piston when the ratio of diameter to stroke is 0.8. | 8        | 3   | 3   | 1   |
| <b>Unit – IV</b>  |                                                                                                                                                                                                                                                                          |          |     |     |     |
| 7.                | a) Explain the working principle of TIG Welding with neat sketch.                                                                                                                                                                                                        | 8        | 2   | 4   | 1   |
|                   | b) List the differences between open belt and crossed belt drives.                                                                                                                                                                                                       | 8        | 2   | 4   | 1   |

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|----|----|-------------------------------------------------------|---|---|---|---|
| 8. | a) | Explain the comparison between welding and soldering. | 8 | 2 | 4 | 1 |
|    | b) | Explain the types of Gears with sketches.             | 8 | 2 | 4 | 1 |

### Unit – V

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|-----|----------------------|------------------------------------------------------|----|---|---|---|
| 9.  | Write a note on      |                                                      |    |   |   |   |
|     | i.                   | Applications of Robots                               |    |   |   |   |
|     | ii.                  | Classification of Robots on different configurations |    |   |   |   |
|     | iii.                 | Robot anatomy with suitable diagrams                 |    |   |   |   |
|     | iv.                  | Open and close loop control system                   | 16 | 2 | 5 | 1 |
| 10. | Explain with figures |                                                      |    |   |   |   |
|     | i.                   | Turning and Facing                                   |    |   |   |   |
|     | ii.                  | Drilling, Boring and Counter Sinking                 |    |   |   |   |
|     | iii.                 | Plane milling and End Milling                        |    |   |   |   |
|     | iv.                  | Elements of automated systems                        | 16 | 2 | 5 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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